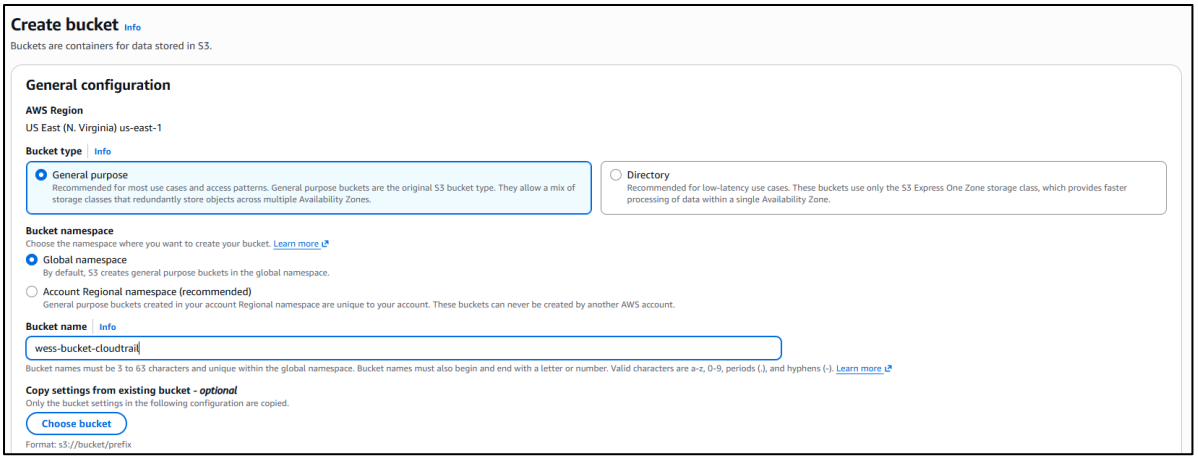
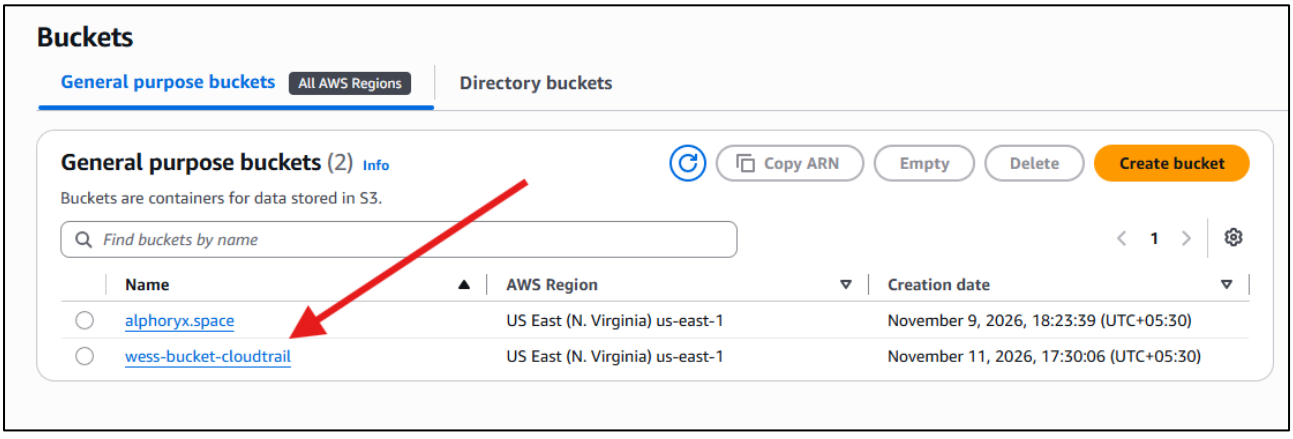
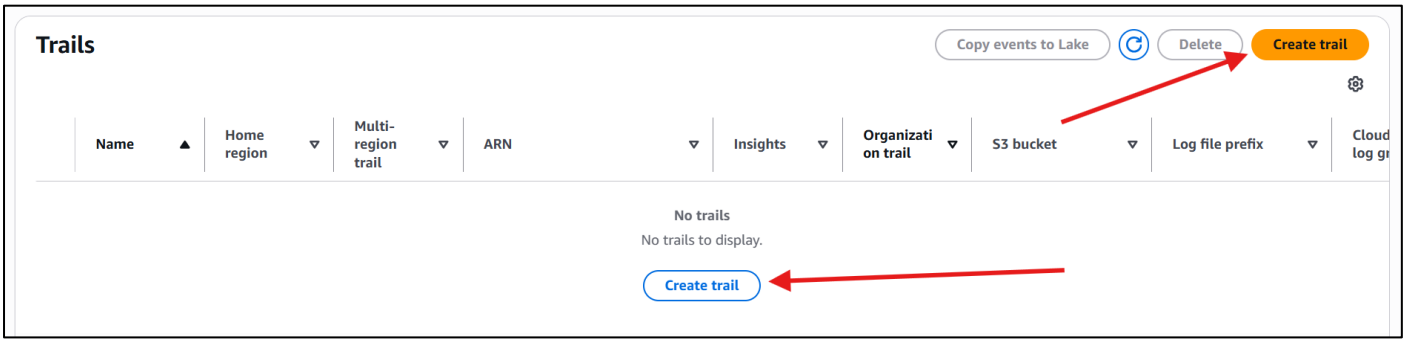
CloudTrail & CloudWatch

1. **Enable CloudTrail monitoring and store the events in S3 and CloudWatch log events.**

Create S3 Bucket → Click Create bucket. → Enter a bucket name. → Keep Block all public access enabled. → Click Create bucket.
Create CloudTrail → Open Trails. → Click Create trail. → Enter the trail name. → Select the existing S3 bucket created earlier. → Enable Log file validation. → Click Next.
Configure Event Monitoring → Select Management events. → Enable Read events. → Enable Write events. → Click Next. → Review the configuration. → Click Create trail.
Configure CloudWatch Logs → Open the created CloudTrail trail. → Navigate to CloudWatch Logs. → Click Edit. → Enable CloudWatch Logs. → Choose existing Log group named / aws-cloudtrail-logs-501794745645-59048cdd → Create or select an IAM role. → Click Save changes.
Verify S3 Logs → Open Amazon S3. → Select your CloudTrail bucket. → Open the AWSLogs folder. → Open your AWS Account ID folder. → Open the CloudTrail folder. → Verify that log files are available.
Verify CloudWatch Logs → Open Amazon CloudWatch. → Navigate to Logs → Log groups → Open / aws-cloudtrail-logs-501794745645-59048cdd → Select the available log stream. → Verify that CloudTrail events are recorded.
Result Below:







Choose trail attributes

General details

A trail created in the console is a multi-region trail. Learn more

Trail name

Enter a display name for your trail.

wess-management-events

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

Enable for all accounts in my organization

To review accounts in your organization, open AWS Organizations. See all accounts

Storage location | Info

- Create new S3 bucket
  Create a bucket to store logs for the trail.
- (•) Use existing S3 bucket
  Choose an existing bucket to store logs for this trail.

Trail log bucket name

Enter a new S3 bucket name and folder (prefix) to store your logs. Bucket names must be globally unique.

wess-bucket-cloudtrail

Browse

CloudWatch Logs - *optional*

Configure CloudWatch Logs to monitor your trail logs and notify you when specific activity occurs. Standard CloudWatch and CloudWatch Logs charges apply. Learn more

CloudWatch Logs | Info

☑ Enabled

Log group | Info

- (•) New
- ( ) Existing

Log group name

aws-cloudtrail-logs-501794745645-59048cdd

1-512 characters. Only letters, numbers, dashes, underscores, forward slashes, and periods are allowed.

Choose log events

Events Info

Record API activity for individual resources, or for all current and future resources in AWS account. Additional charges apply

Event type

Choose the type of events that you want to log.

- ☑ Management events
  Capture management operations performed on your AWS resources.
- ☐ Data events
  Log the resource operations performed on or within a resource.
- ☐ Insights events
  Identify unusual activity, errors, or user behavior in your account.
- ☐ Network activity events
  Network activity events provide information about resource operations performed on a resource within a virtual private cloud endpoint.

Trail successfully created

Trails

Copy events to Lake | Delete | Create trail

Name	Home region	Multi-region trail	ARN	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group	Sta
wess-management-events	US East (N. Virginia)	Yes	arn:aws:cloudtrail:us-east-1:501794745645:trail/wess-management-events	Disabled	No	wess-bucket-cloudtrail	-	arn:aws:logs:us-east-1:501794745645:log-group:aws-cloudtrail-logs-501794745645-59048cdd:*	ng

wess-bucket-cloudtrail Info

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems | Access Points

Objects (1)

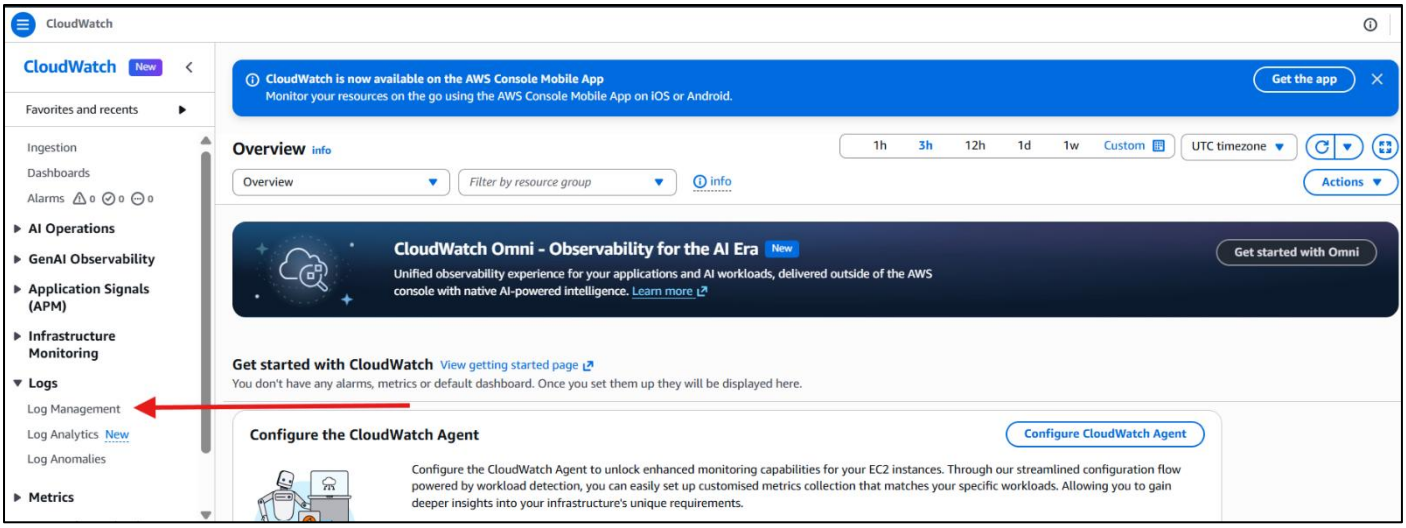
Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

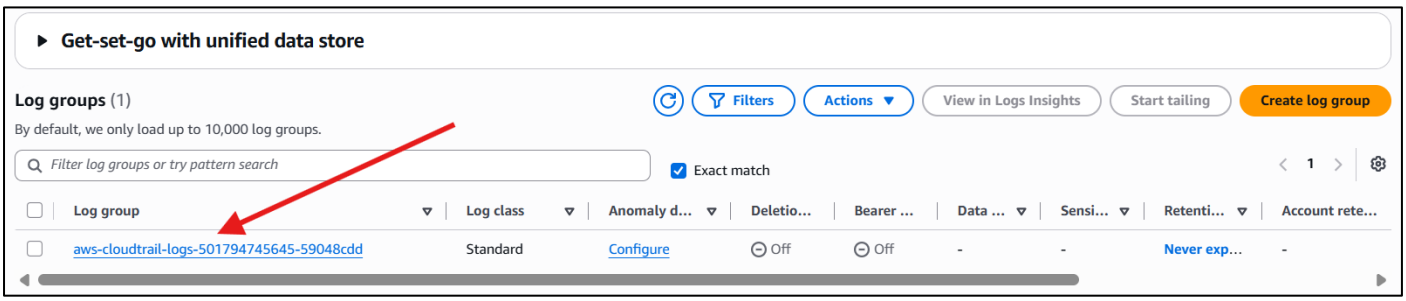
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

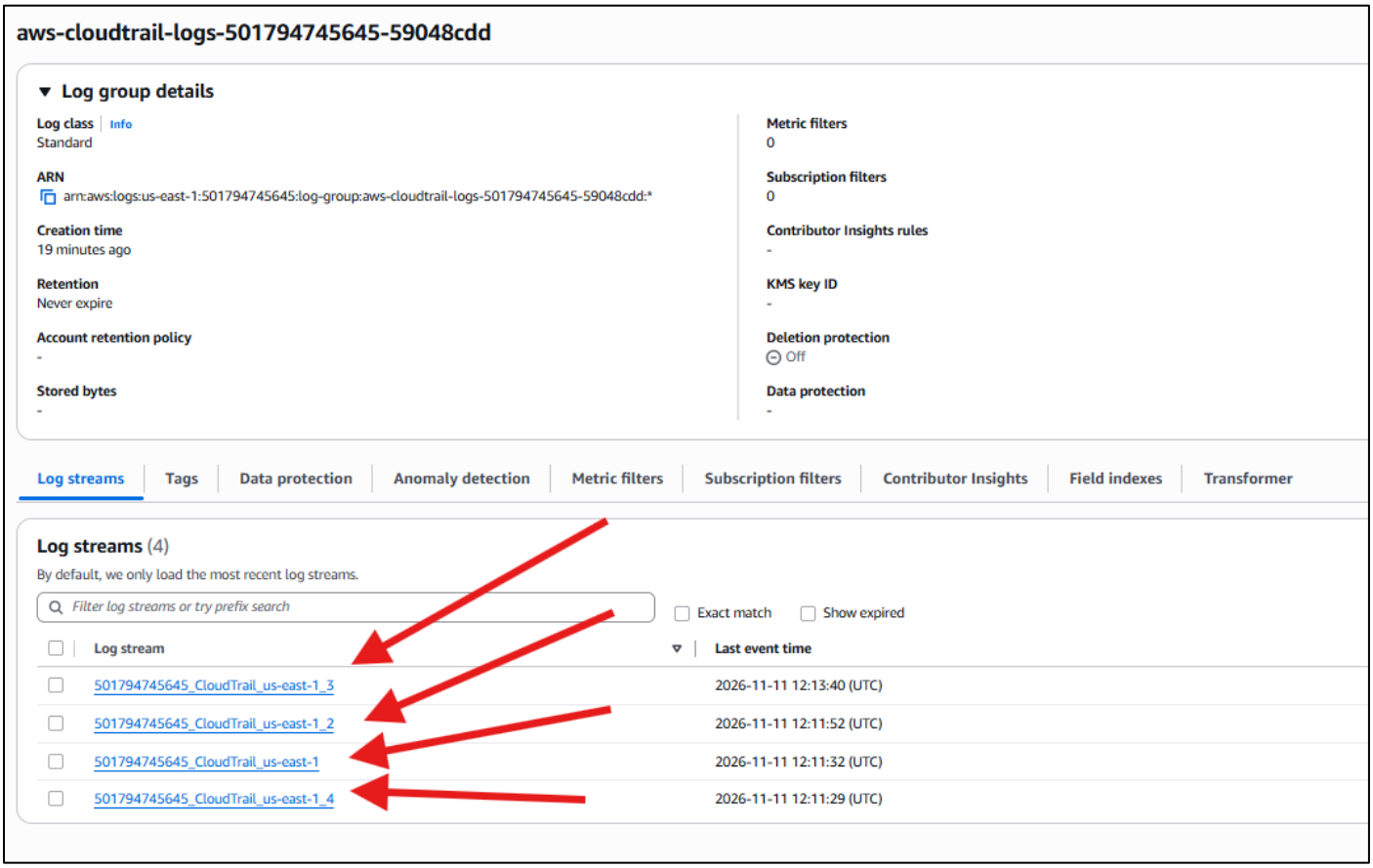
Find objects by prefix

1

Name	Type	Last modified	Size	Storage class
AWSLogs/	Folder	-	-	-


CloudWatch
CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
Log Analytics New
Log Anomalies
Metrics
CloudWatch is now available on the AWS Console Mobile App
Monitor your resources on the go using the AWS Console Mobile App on iOS or Android.
Get the app
Overview info
1h 3h 12h 1d 1w Custom
UTC timezone
Actions
Filter by resource group
CloudWatch Omni - Observability for the AI Era New
Unified observability experience for your applications and AI workloads, delivered outside of the AWS console with native AI-powered intelligence. Learn more
Get started with Omni
Get started with CloudWatch View getting started page
You don't have any alarms, metrics or default dashboard. Once you set them up they will be displayed here.
Configure the CloudWatch Agent
Configure CloudWatch Agent
Configure the CloudWatch Agent to unlock enhanced monitoring capabilities for your EC2 instances. Through our streamlined configuration flow powered by workload detection, you can easily set up customised metrics collection that matches your specific workloads. Allowing you to gain deeper insights into your infrastructure's unique requirements.

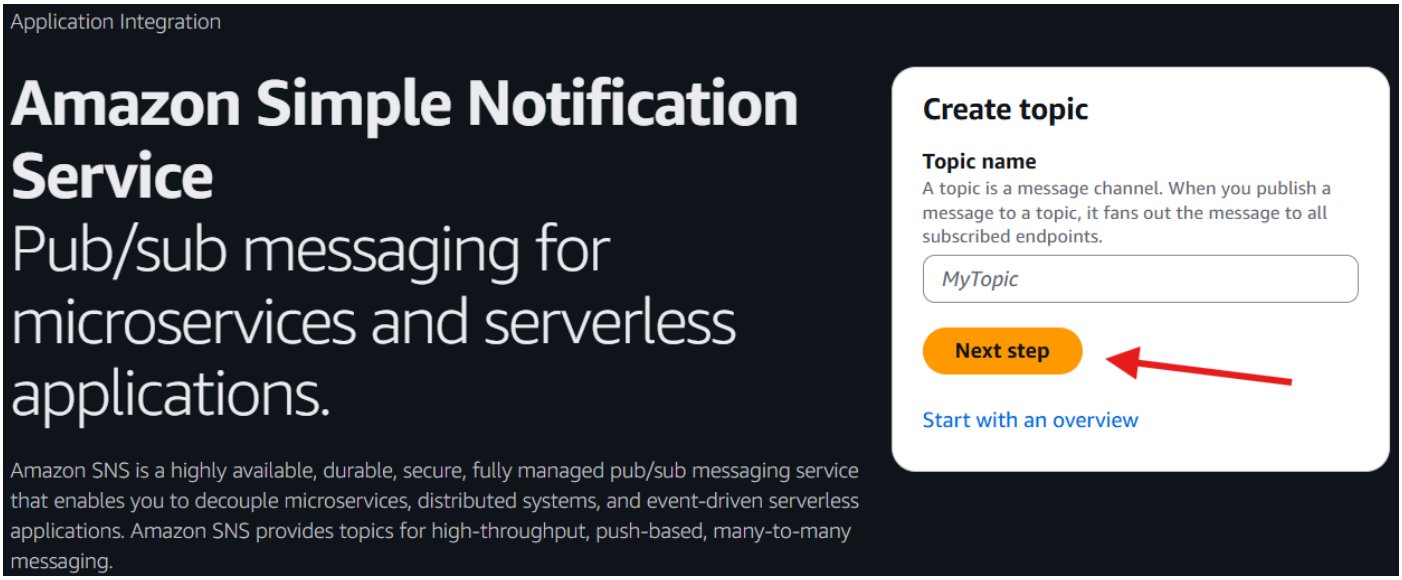

Get-set-go with unified data store
Log groups (1)
By default, we only load up to 10,000 log groups.
Filters
Actions
View in Logs Insights
Start tailing
Create log group
Filter log groups or try pattern search
Exact match
Log group
Log class
Anomaly d...
Deletio...
Bearer ...
Data ...
Sensi...
Retenti...
Account rete...
aws-cloudtrail-logs-501794745645-59048cdd
Standard
Configure
Off
Off
Never exp...

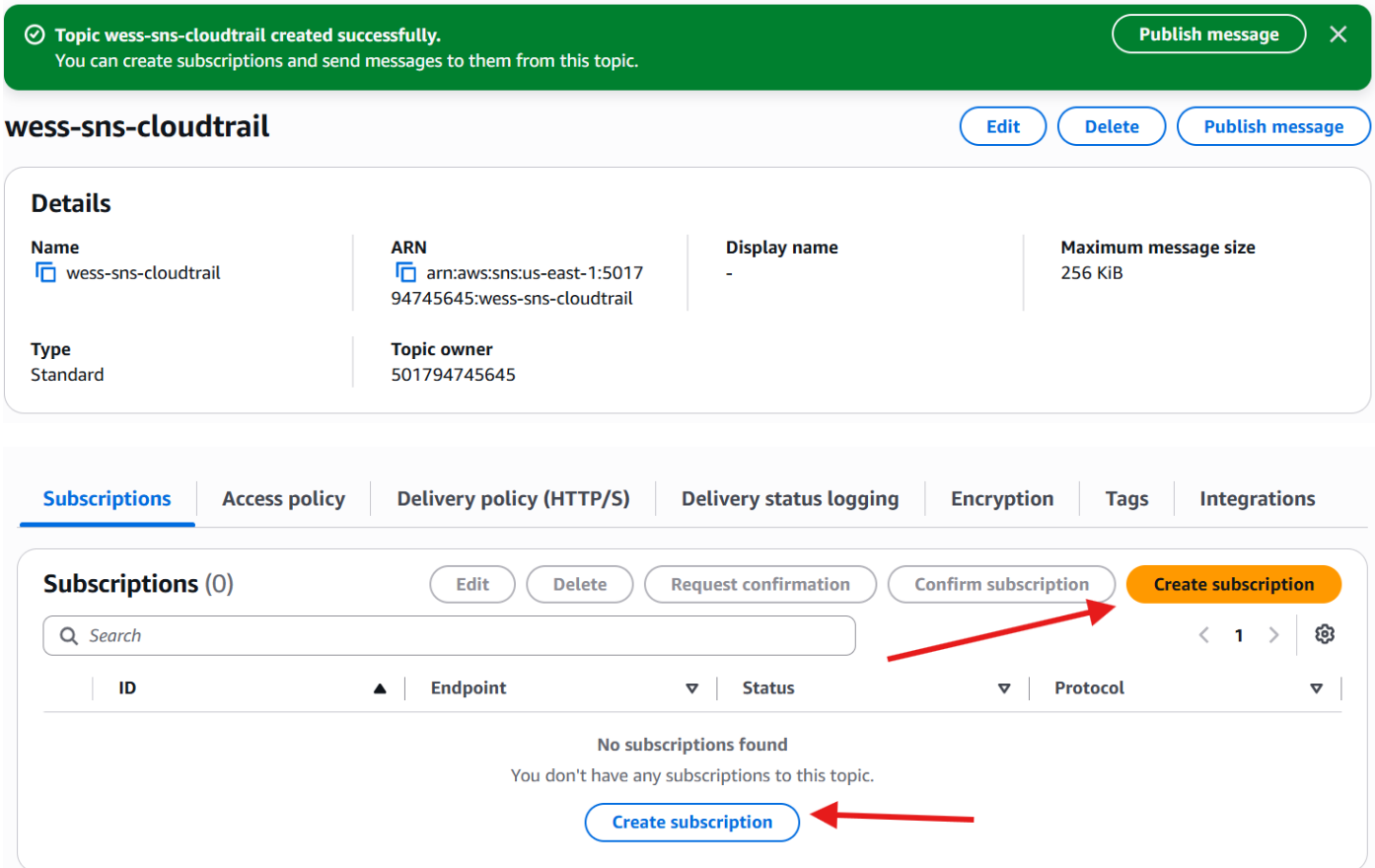

aws-cloudtrail-logs-501794745645-59048cdd
Log group details
Log class Info
Standard
ARN
arn:aws:logs:us-east-1:501794745645:log-group:aws-cloudtrail-logs-501794745645-59048cdd:*
Creation time
19 minutes ago
Retention
Never expire
Account retention policy
-
Stored bytes
-
Metric filters
0
Subscription filters
0
Contributor Insights rules
-
KMS key ID
-
Deletion protection
Off
Data protection
-
Log streams
Tags
Data protection
Anomaly detection
Metric filters
Subscription filters
Contributor Insights
Field indexes
Transformer
Log streams (4)
By default, we only load the most recent log streams.
Filter log streams or try prefix search
Exact match
Show expired
Log stream
Last event time
501794745645_CloudTrail_us-east-1_3
2026-11-11 12:13:40 (UTC)
501794745645_CloudTrail_us-east-1_2
2026-11-11 12:11:52 (UTC)
501794745645_CloudTrail_us-east-1
2026-11-11 12:11:32 (UTC)
501794745645_CloudTrail_us-east-1_4
2026-11-11 12:11:29 (UTC)

2. Enable SNS for CloudTrail to send alerts via email

Create an SNS Topic → Search for Amazon SNS. → Select Topics.→ Click Create topic. → Select Standard as the topic type. → Enter the topic name: wess-sns-cloudtrail.→ Click Create topic.
Create Email Subscription → Open the CloudTrail-Alerts topic. → Click Create subscription. → Select Email as the protocol. → Enter your email address in the Endpoint field. → Click Create subscription. → Open your email inbox. → Find the AWS SNS confirmation email. → Click Confirm subscription. → Verify that the subscription status shows Confirmed.
Enable SNS Notifications in CloudTrail → Open AWS CloudTrail. → Select Trails. → Open your existing CloudTrail trail. → Click Edit. → Find the SNS notification delivery section. → Enable SNS notifications. → Select the existing SNS topic wess-sns-cloudtrail. → Save the changes.

Results Below:





Create subscription

Details

Topic ARN

arn:aws:sns:us-east-1:501794745645:wess-sns-cloudtrail

Protocol

The type of endpoint to subscribe

Email

Endpoint

An email address that can receive notifications from Amazon SNS.

viquar44@gmail.com

After your subscription is created, you must confirm it. Info

▸ **Subscription filter policy - *optional*** Info

This policy filters the messages that a subscriber receives.

▸ **Redrive policy (dead-letter queue) - *optional*** Info

Send undeliverable messages to a dead-letter queue.

Cancel | Create subscription

AWS Notification - Subscription Confirmation

Inbox × | Convert to PDF

AWS Notifications <no-reply@sns.amazonaws.com>

to me

7:07 PM (0 minutes ago)

You have chosen to subscribe to the topic:
arn:aws:sns:us-east-1:501794745645:wess-sns-cloudtrail

To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
Confirm subscription

Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to sns-opt-out

aws

Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:
arn:aws:sns:us-east-1:501794745645:wess-sns-cloudtrail:8c746bf4-dde0-49f4-a1ab-c48fe825039f

If it was not your intention to subscribe, click here to unsubscribe.

▼ **Additional settings**

Log file validation | Info

☑ Enabled

Recursive logging | Info

☑ Enabled

SNS notification delivery | Info

☑ Enabled

Create a new SNS topic

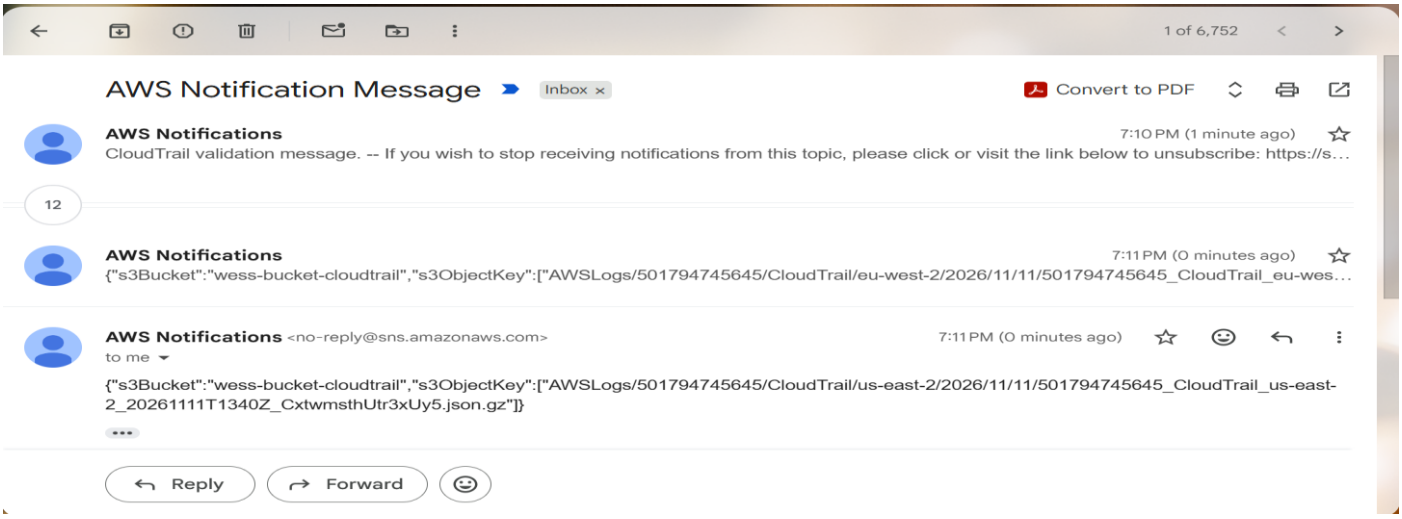
○ New

◉ Existing

SNS topic

arn:aws:sns:us-east-1:501794745645:wess-sns-cloudtrail

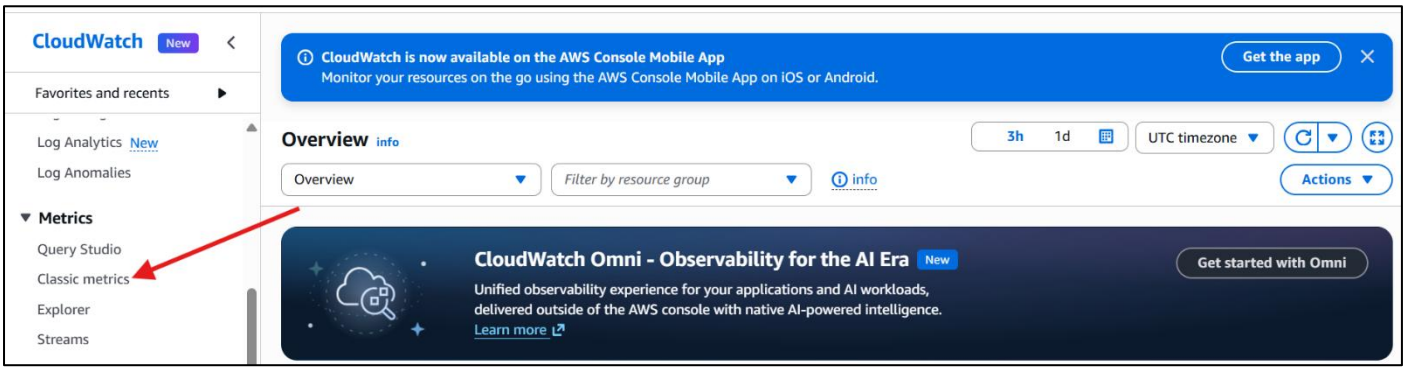
Cancel | Save changes

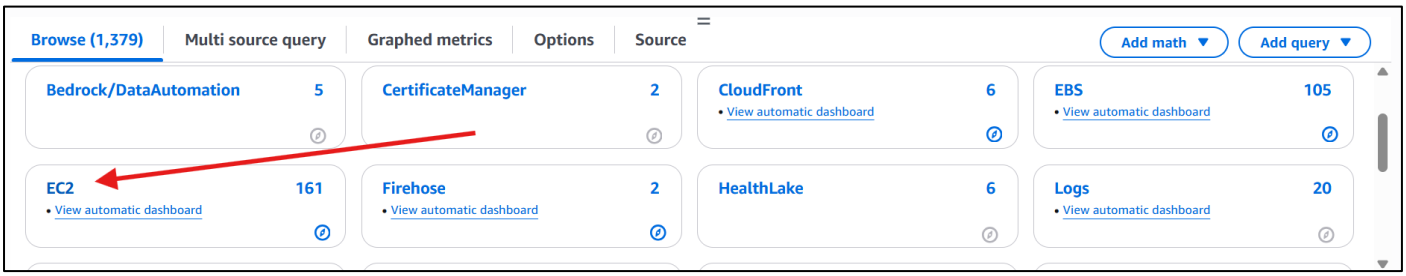


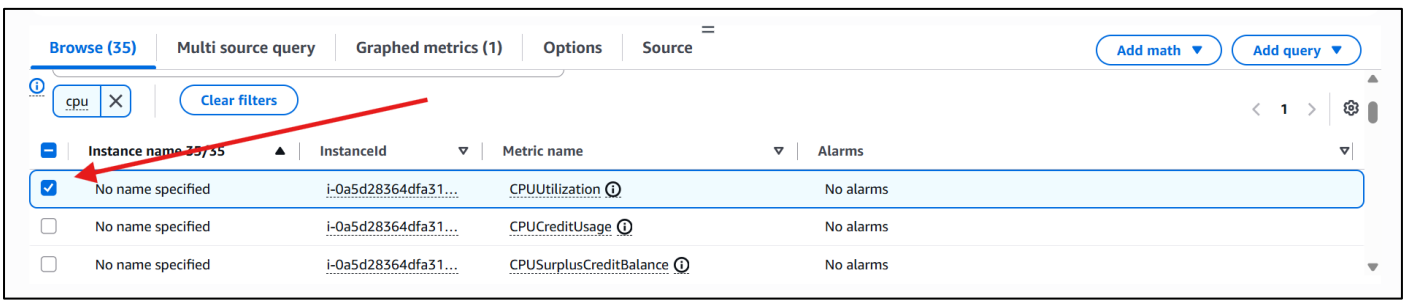
3. Configure CloudWatch monitoring and record the CPU utilization and other metrics of EC2.

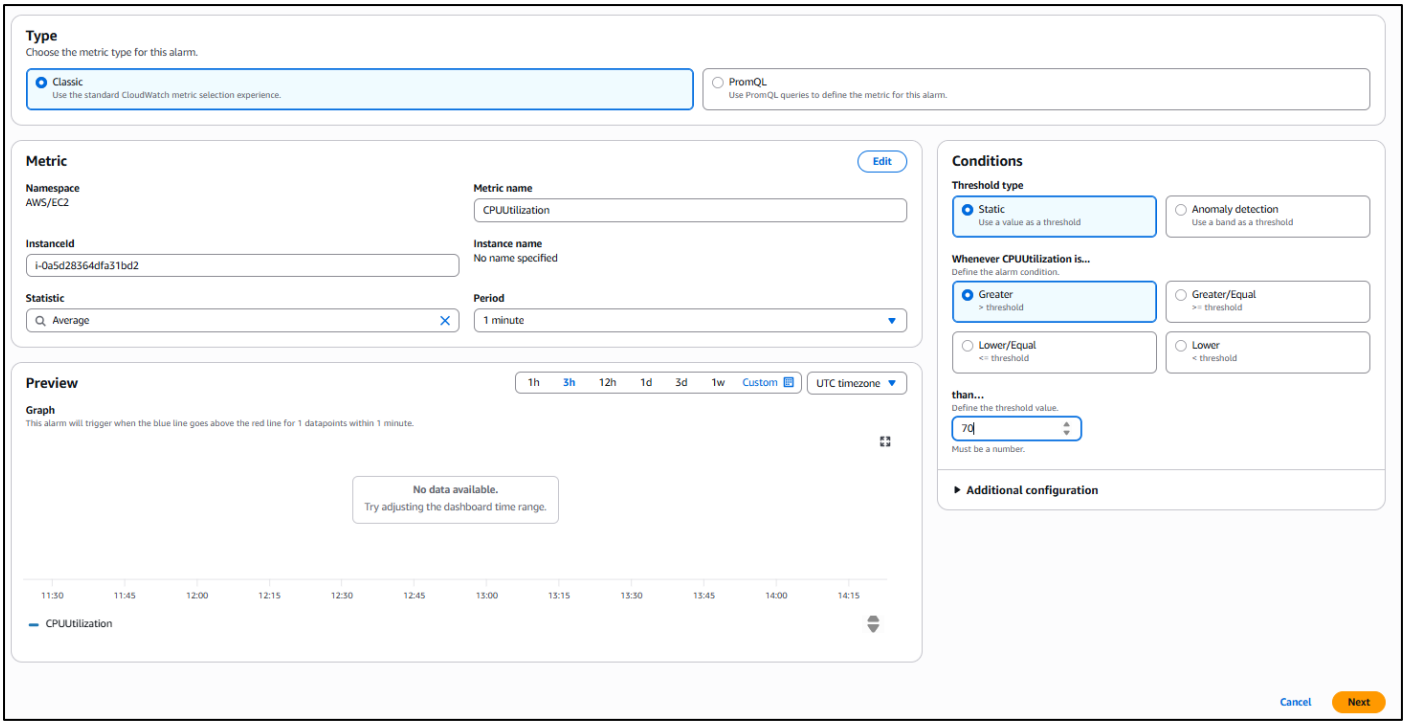
Open CloudWatch → Open CloudWatch. → Select Metrics from the left navigation panel. → Click All metrics.

Monitor EC2 CPU Utilization → Select EC2. → Select Per-Instance Metrics. → Find your EC2 instance using its Instance ID. → Select the CPUUtilization metric. → View the CPU utilization graph. → Select the required time range to monitor CPU usage.



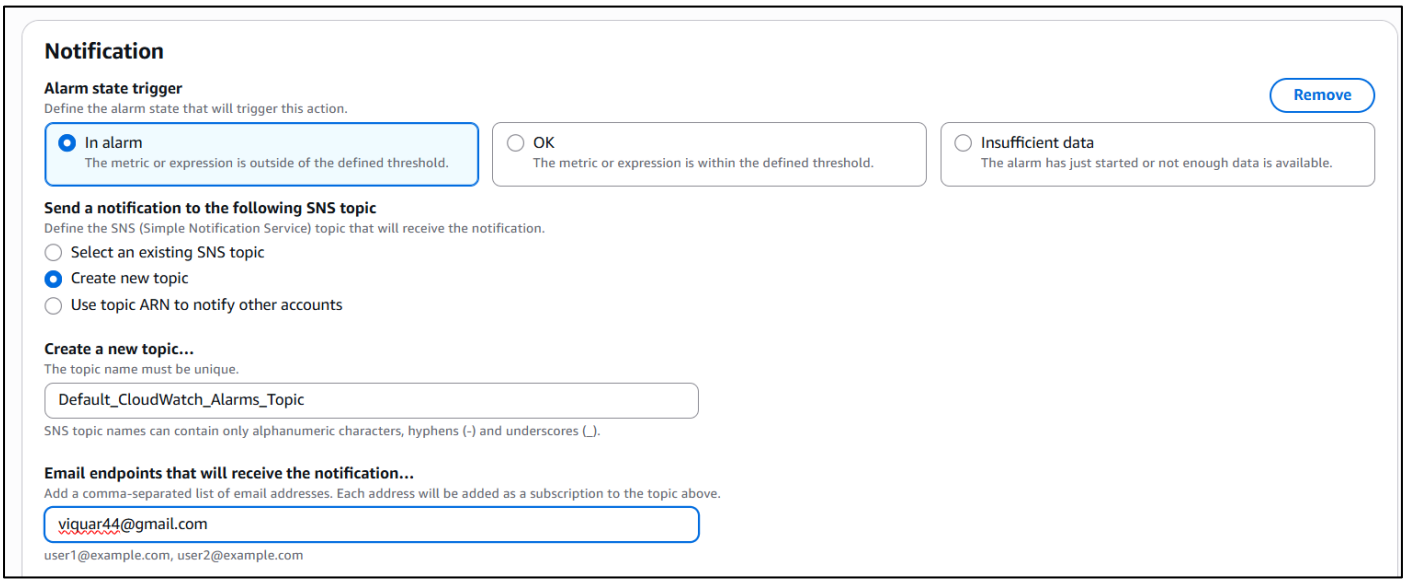


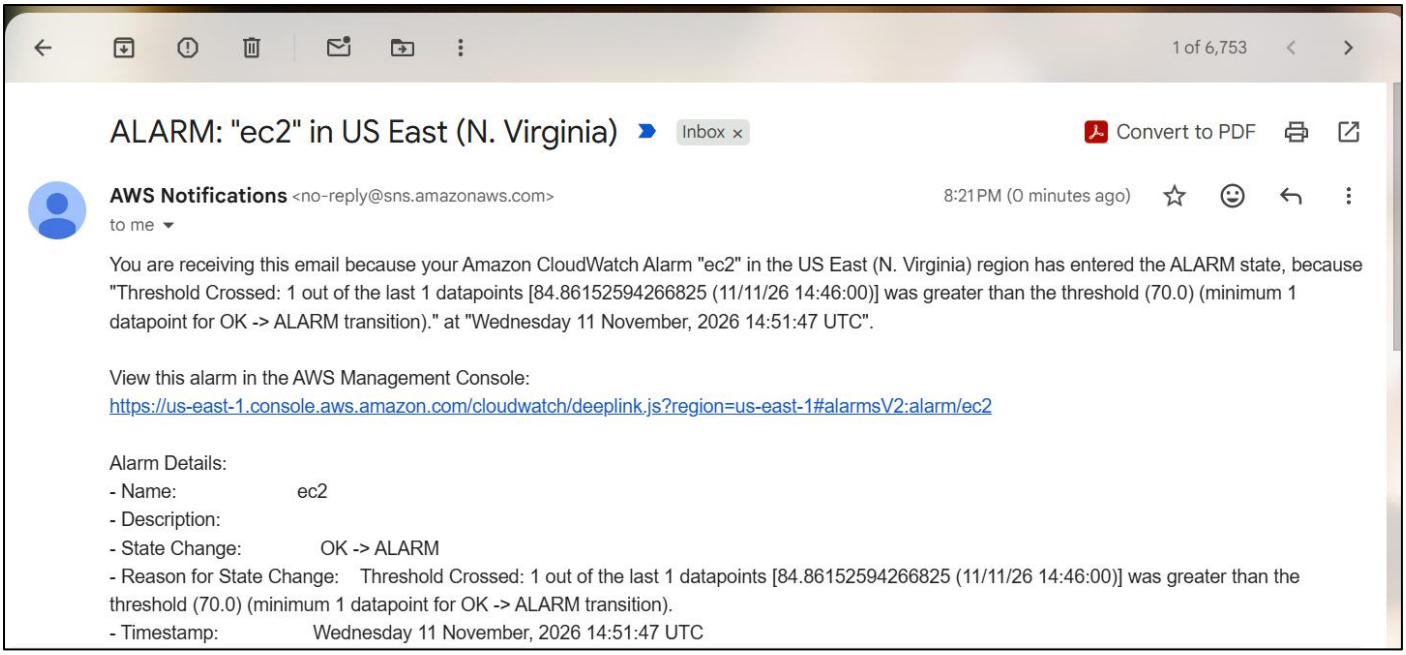




4. Create one alarm to send an alert to email if the CPU utilization is more than 70 percent.

Open AWS Management Console. → Go to Amazon CloudWatch. → Navigate to Alarms → All alarms. → Click Create alarm. → Click Select metric. → Choose EC2 → Per-Instance Metrics. → Select your EC2 instance's CPUUtilization metric. → Click Select metric. → Set the Statistic to Average. → Set the Period to 5 minutes. → Under Conditions, select Greater than 70. → Click Next. → Under Notification, select In alarm. Select Create new SNS topic. → Enter the topic name EC2-CPU-Alert. → Select Email as the notification protocol. → Enter your email address. → Click Create topic. → Click Next. → Enter the alarm name EC2-CPU-Above-70. → Click Next → Create alarm. → Open your email inbox. → Confirm the SNS subscription. → Verify that the alarm status changes to OK when CPU utilization is below 70%. → Verify that an email alert is received when CPU utilization exceeds 70%.





1 of 6,753

ALARM: "ec2" in US East (N. Virginia) Inbox ×

Convert to PDF

AWS Notifications <no-reply@sns.amazonaws.com>

8:21 PM (0 minutes ago)

to me

You are receiving this email because your Amazon CloudWatch Alarm "ec2" in the US East (N. Virginia) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [84.86152594266825 (11/11/26 14:46:00)] was greater than the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Wednesday 11 November, 2026 14:51:47 UTC".

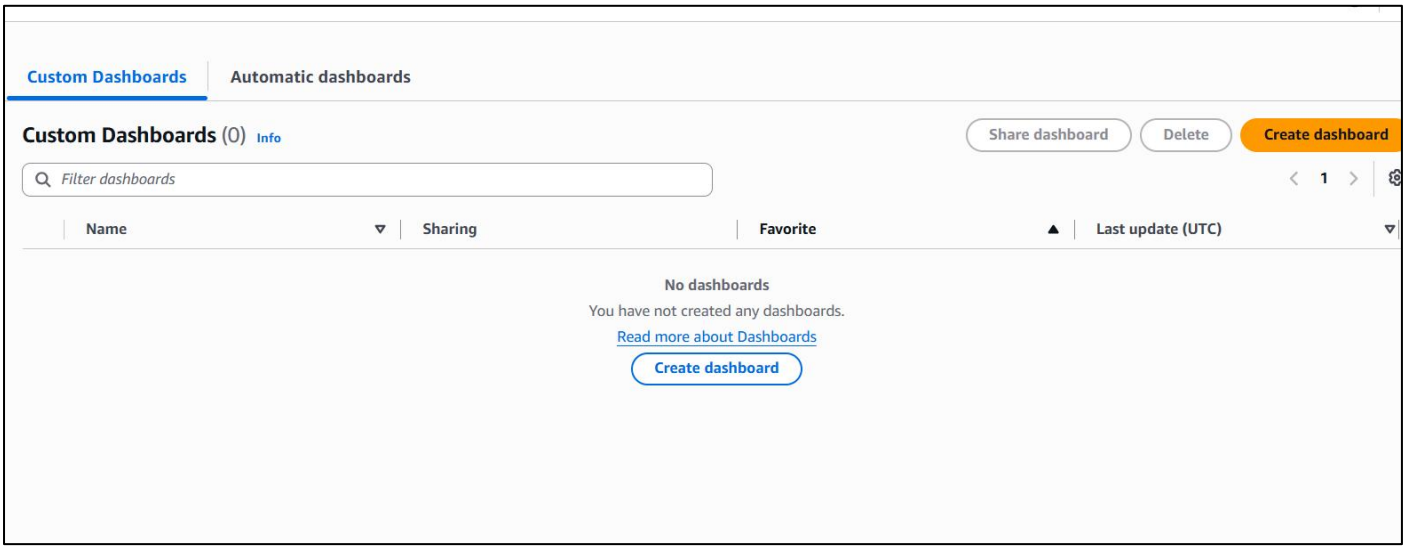
View this alarm in the AWS Management Console:
https://us-east-1.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-1#alarmsV2:alarm/ec2

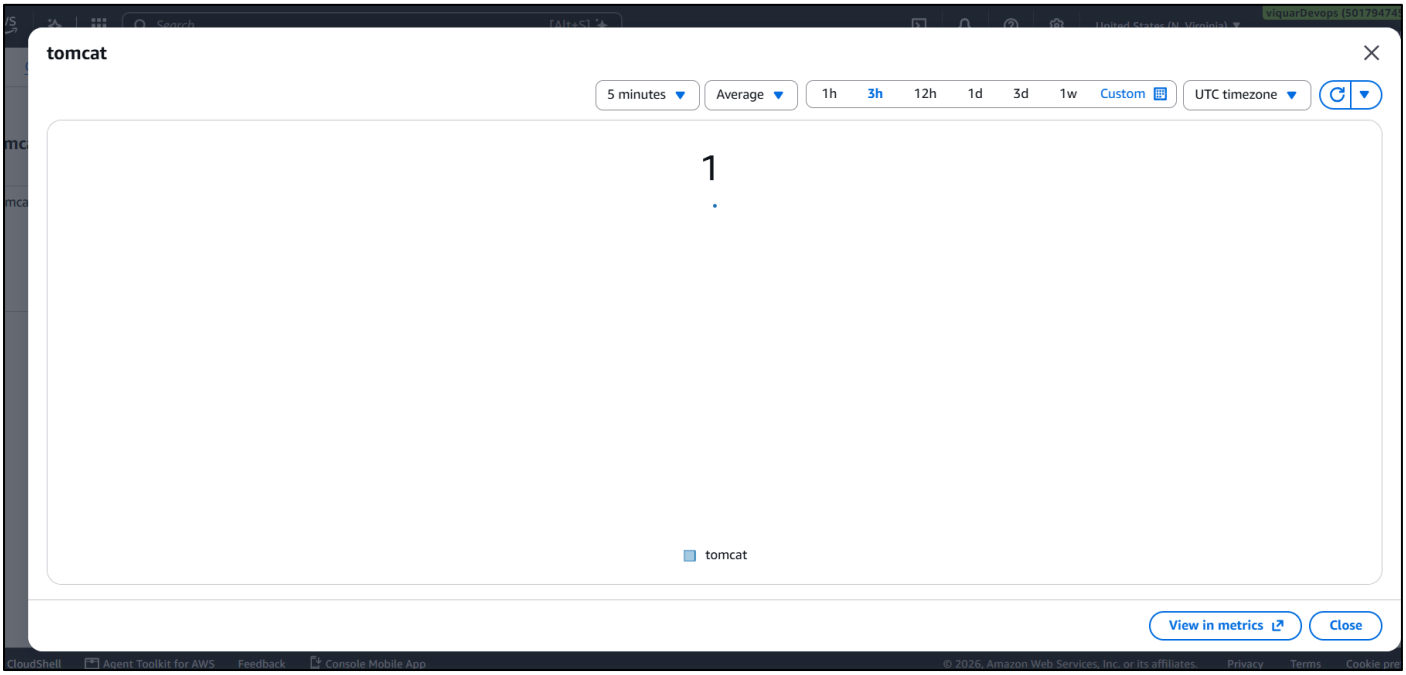
Alarm Details:
- Name: ec2
- Description:
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [84.86152594266825 (11/11/26 14:46:00)] was greater than the threshold (70.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Wednesday 11 November, 2026 14:51:47 UTC

5. Create a Dashboard and monitor the Tomcat service whether it is running or not and send the alert.

Connected to the EC2 instance using SSH. → Downloaded the Apache Tomcat package using the wget command. → Extracted the downloaded Tomcat archive using the tar command. → Installed and configured Tomcat using the .sh installation script. → Created a Bash script to monitor the Tomcat service status.
→ Configured the script to send Tomcat monitoring data to Amazon CloudWatch.
→ Created a CloudWatch Dashboard to monitor whether the Tomcat service is running or stopped. → Created a CloudWatch Alarm to detect when the Tomcat service goes down. → Configured an Amazon SNS email notification to receive alerts.

Tested the monitoring by stopping and restarting Tomcat and verified the dashboard status and email alerts.





6. Create a Dashboard and monitor the Nginx service to send the alert if Nginx is not running.

Installed and configured Nginx on the EC2 instance. → Created a Bash script to monitor the Nginx service status. → Configured the script to send monitoring data to Amazon CloudWatch. → Created a CloudWatch Dashboard to monitor whether Nginx is running or stopped.
Created a CloudWatch Alarm to detect if the Nginx service stops. → Configured Amazon SNS to send email alerts.

Tested the setup by stopping and restarting Nginx and verified the dashboard status and email notifications.

```
[root@ip-172-31-16-171 ~]# systemctl status nginx
○ nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
     Active: inactive (dead)
[root@ip-172-31-16-171 ~]# systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.
[root@ip-172-31-16-171 ~]# systemctl start nginx
[root@ip-172-31-16-171 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-11-12 06:30:00 UTC; 6s ago
    Process: 56527 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
    Process: 56529 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
    Process: 56544 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
   Main PID: 56555 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 81ms
     CGroup: /system.slice/nginx.service
             ├─56555 "nginx: master process /usr/sbin/nginx"
             ├─56557 "nginx: worker process"
             └─56559 "nginx: worker process"

Nov 12 06:29:59 ip-172-31-16-171.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reve
Nov 12 06:30:00 ip-172-31-16-171.ec2.internal nginx[56529]: nginx: the configuration file /etc/nginx/nginx
Nov 12 06:30:00 ip-172-31-16-171.ec2.internal nginx[56529]: nginx: configuration file /etc/nginx/nginx.con
Nov 12 06:30:00 ip-172-31-16-171.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and rever
[root@ip-172-31-16-171 ~]#
```

root@ip-172-31-16-171:~

```
#!/bin/bash
# Configuration
NAMESPACE="Custom/NGINX"
METRIC_NAME="NginxStatus"
REGION="us-east-1"  # Change this to your preferred AWS region
# AWS CLI command for sending metrics
AWS_CLI_CMD="aws cloudwatch put-metric-data --namespace $NAMESPACE --metric-
# Check if NGINX is running
if pgrep nginx > /dev/null; then
    STATUS=1
else
    STATUS=0
fi
# Send the metric to CloudWatch
$AWS_CLI_CMD --value $STATUS --unit Count
# Logging for debugging
echo "NginxStatus: $STATUS" >> /var/log/nginx_monitor.log
~
~
~
```

